

# CSE 813: Machine Learning

## Week 1, Part 1 — Introduction to Machine Learning

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# Course & Lecture Information

## Course Details

**Code:** CSE 813  
**Credits:** 3 (Theory)  
**Pre-req:** CSE 221 (Data Structures),  
CSE 225 (Algorithms),  
CSE 235 (Probability & Stats)

## Today

**Week / Part:** 1 of 13 | Part 1 of 2  
**CLO:** CLO1  
**PLO:** PLO(a)

## CLO1

*“Understand the fundamental concepts, paradigms, and theoretical foundations of machine learning”*

## Part 1 Covers

- What is Machine Learning?
- The formal definition (Mitchell, 1997)
- The three learning paradigms
- The ML pipeline
- Key concepts: generalization, overfitting, underfitting, inductive bias
- Ivy League alignment
- Student assessment pool

# Lecture Agenda

- 1 The Machine Learning Revolution
- 2 Formal Definition (Mitchell, 1997)
- 3 The Three Learning Paradigms
- 4 The Machine Learning Pipeline
- 5 Key Concepts: Overfitting, Underfitting, Generalization
- 6 Ivy League Alignment
- 7 Student Assessment Pool

# The Machine Learning Revolution

# Why Machine Learning Now?

## The AI Transformation

Machine learning has moved from academic research to **every aspect of modern life**:

- **Consumer**: Netflix recommendations, Amazon product suggestions, Spotify Discover Weekly
- **Communication**: Gmail spam filter, Google Translate, voice assistants (Siri, Alexa)
- **Transportation**: Tesla Autopilot, Uber surge pricing, Waze routing
- **Healthcare**: Disease diagnosis from medical imaging, drug discovery (AlphaFold)
- **Finance**: Fraud detection, credit scoring, algorithmic trading

## Market Context

Global AI market: projected **\$1.8 trillion by 2030** (industry forecasts)  
ML/AI roles: among the **fastest-growing job categories** worldwide (WEF *Future of Jobs*)  
Bangladesh: AI/ML a **fast-growing segment** of national ICT exports

## Why Now? (4 Drivers)

- **Data explosion**: global datasphere  $\approx$  175 ZB (IDC projection, 2025)
- **Compute power**: GPUs, TPUs, cloud
- **Algorithms**: Deep learning breakthroughs
- **Open source**: TensorFlow, PyTorch, scikit-learn democratised access

## Timeline of AI & ML Milestones

|      |                                                          |
|------|----------------------------------------------------------|
| 1959 | Samuel – self-improving checkers player                  |
| 1986 | Backpropagation popularised (Rumelhart et al.)           |
| 1997 | Deep Blue beats Kasparov ( <i>search, not learning</i> ) |
| 2012 | AlexNet – ImageNet breakthrough                          |
| 2016 | AlphaGo beats Lee Sedol                                  |
| 2020 | AlphaFold solves protein structure prediction            |
| 2022 | ChatGPT – LLMs reach the public                          |
| 2024 | Nobel Prizes recognise ML research                       |

# What Makes ML Different from Traditional Programming?

## Traditional Programming

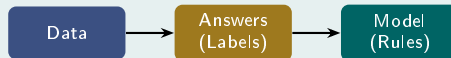


**Human writes rules**; machine executes them.

**Example:** Spam filter with hand-coded rules

- IF sender NOT in contacts THEN flag
- IF contains "lottery" THEN flag
- (1000+ rules maintained by engineers)

## Machine Learning



**Machine learns rules** from data + answers. **Example:**

ML spam filter

- Train on 100,000 labeled emails
- Model learns patterns automatically
- No human-written rules needed
- Achieves 99.9% accuracy

## The Paradigm Shift

Traditional programming: **explicit** knowledge encoding.

Machine learning: **implicit** knowledge discovery from data.

Formal Definition (Mitchell, 1997)

# The Canonical Definition: Tom Mitchell (1997)

## Full Citation

Mitchell, T. M. (1997). **Machine Learning**. McGraw-Hill. *The first comprehensive textbook; still a standard reference.*

## The Definition

*“A computer program is said to **learn** from experience **E** with respect to some class of tasks **T** and performance measure **P**, if its performance at tasks in **T**, as measured by **P**, **improves with experience E.**”* —Tom Mitchell, 1997

## Deconstructing the Definition: Spam Filter

- **Task T**: Classify emails as spam or not spam
- **Performance P**: Accuracy (% correctly classified)
- **Experience E**: 100,000 labeled emails (spam/ham)

*Performance improves as the model sees more labeled emails.*

## Why This Definition Matters

- Provides a **rigorous framework** for what “learning” means
- Every ML problem can be specified as **T, P, E**
- Forces practitioners to be **precise** about the task, the success metric, and the available data
- Enables **comparison** across algorithms

## The Earlier, Informal Definition

*“The field of study that gives computers the ability to learn without being explicitly programmed.”*  
—attributed to Arthur Samuel, 1959

*Widely quoted, but a later paraphrase—it does not appear verbatim in Samuel’s 1959 paper (examined in Part 2).*



# Applying Mitchell's Framework: Examples

| Problem                | Task (T)                    | Experience (E)            | Performance (P)        |
|------------------------|-----------------------------|---------------------------|------------------------|
| Spam detection         | Classify email as spam/ham  | 1M labeled emails         | Accuracy, F1-score     |
| House price prediction | Predict price (regression)  | Historical sales records  | RMSE, $R^2$            |
| Credit scoring         | Predict default risk        | Loan repayment history    | ROC-AUC                |
| Product recommendation | Rank products for a user    | User purchase history     | Click-through rate     |
| Image classification   | Identify object in image    | Labeled image dataset     | Top-1 / Top-5 accuracy |
| Fraud detection        | Flag fraudulent transaction | Past labeled transactions | Precision / Recall     |

## The Critical Insight

Without a clearly defined T, P, E, you are not doing machine learning—you are just writing code. The definition keeps us honest about the goal.

## The Three Learning Paradigms

# Supervised Learning: Learning with Labeled Data

## Definition

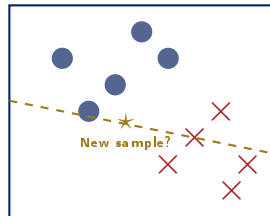
**Supervised learning** learns a mapping  $f : x \mapsto y$  from labeled examples  $\{(x_i, y_i)\}_{i=1}^n$ , then predicts  $y$  for unseen  $x$ .

## Two Main Sub-types

- **Classification:** discrete class label
  - Binary (spam vs. ham), multi-class (digits 0–9), multi-label (image tags)
- **Regression:** continuous value
  - House price, temperature, demand forecasting

## Local Example

**Mobile-money credit scoring** (e.g. bKash): predict loan default risk—binary classification trained on past transactions with known repayment outcomes.



Training Data (labeled)

## Key Algorithms

Linear Regression, Logistic Regression, SVM, Decision Trees, Random Forest, Neural Networks

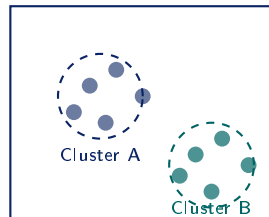
# Unsupervised Learning: Finding Hidden Structure

## Definition

**Unsupervised learning** discovers hidden structure in **unlabeled** data  $\{x_i\}_{i=1}^n$ —no target values are provided.

## Main Sub-types

- **Clustering:** Group similar data points together
  - Customer segmentation
  - Image compression
  - Document organisation
- **Dimensionality Reduction:** Find lower-dimensional representation
  - PCA, t-SNE, UMAP
  - Data visualization
  - Feature extraction
- **Anomaly Detection:** Identify unusual data points
  - Fraud detection, equipment failure prediction



Unlabeled Data

## Key Algorithms

K-Means, Hierarchical Clustering, DBSCAN, PCA, t-SNE

# Reinforcement Learning: Learning from Actions & Rewards

## Definition

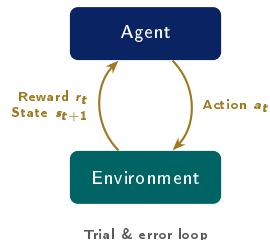
**Reinforcement learning** learns a **policy** by **trial and error**, maximising cumulative **reward**. An agent acts in an **environment** and receives rewards or penalties.

## Key Components

- **Agent**: the learner / decision-maker
- **Environment**: what the agent acts upon
- **State**: current situation; **Action**: what it may do
- **Reward**: scalar feedback (positive or negative)
- **Policy**: strategy mapping states to actions

## Game Playing (AlphaGo)

State = board position; Actions = possible moves; Reward = win/loss.  
AlphaGo learned by playing millions of games against itself.



## Key Applications

- Game playing (AlphaGo, chess, poker)
- Robotics, control, autonomous vehicles
- Resource management, sequential recommendation

# The Three Paradigms: Comparison

| Dimension     | Supervised              | Unsupervised                  | Reinforcement                                   |
|---------------|-------------------------|-------------------------------|-------------------------------------------------|
| Data          | Labeled $(x, y)$ pairs  | Unlabeled $x$ only            | Sequential states, actions, rewards             |
| Goal          | Predict $y$ from $x$    | Discover hidden structure     | Maximize cumulative reward                      |
| Feedback      | Immediate (label known) | None (no ground truth)        | Delayed and evaluative                          |
| Key challenge | Generalization          | Evaluation / interpretability | Credit assignment; exploration vs. exploitation |
| Example       | Spam filter             | Customer segmentation         | AlphaGo                                         |

## Beyond the Three

**Semi-supervised:** a few labels + much unlabeled data.

**Self-supervised:** labels generated from the data itself (masked-token prediction)—the engine behind modern LLMs.

## Course Focus

CSE 813 covers **supervised and unsupervised learning** in depth (Weeks 3–11). Reinforcement learning is treated **conceptually only**—it is defined and contrasted here, but not developed or assessed algorithmically.

# The Machine Learning Pipeline

# The End-to-End ML Pipeline



| Stage                  | Description                                                             |
|------------------------|-------------------------------------------------------------------------|
| 1. Data collection     | Gather raw data from sources (databases, APIs, sensors, surveys)        |
| 2. Data preprocessing  | Clean data: handle missing values, remove duplicates, scale/normalise   |
| 3. Feature engineering | Transform raw data into features the model can use                      |
| 4. Model selection     | Choose algorithm(s) and tune hyperparameters via <b>validation</b> data |
| 5. Model training      | Learn parameters from the training data                                 |
| 6. Model evaluation    | Assess performance on <b>held-out test</b> data, once                   |
| 7. Model deployment    | Serve in production, monitor for <b>data/concept drift</b> , retrain    |

## Key Insight

In real-world ML, **stages 1–3 (data work)** typically consume the majority of project time (practitioner surveys report  $\sim 80\%$ ). The modelling steps are the smallest part—contrary to public perception. (Cf. *Sculley et al., 2015, “Hidden Technical Debt in ML Systems”, Week 12.*)

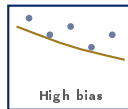


Key Concepts: Overfitting, Underfitting, Generalization

# The Central Challenge: Generalization

## Generalization

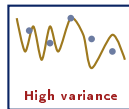
The ability of a model to perform well on **new, unseen data**—not merely on the data it was trained on. This, not training accuracy, is **the goal** of machine learning.



Underfitting



Just Right



Overfitting

## How We Measure It: Data Splits

- **Training set** (~60–80%): fit model parameters
- **Validation set** (~10–20%): tune hyperparameters, select models
- **Test set** (~10–20%): **touched once**, for the final unbiased estimate

*Generalization gap* = test error – training error.

Model complexity → (left: too simple; right: too complex)

## The Goldilocks Problem

A model must be **complex enough** to capture the pattern, yet **simple enough** to generalise. This tension is formalised next as the **bias–variance trade-off**.

## Golden Rule

Never make a modelling decision using the test set—that leaks information and inflates reported performance.

# Overfitting vs. Underfitting: A Closer Look

| Dimension        | Overfitting                              | Underfitting                                  |
|------------------|------------------------------------------|-----------------------------------------------|
| Definition       | Model learns noise, not signal           | Model fails to capture the underlying pattern |
| Training error   | Very low                                 | High                                          |
| Test error       | High                                     | High                                          |
| Model complexity | Too high                                 | Too low                                       |
| Dominant term    | Variance                                 | Bias                                          |
| Cause            | Too many parameters, insufficient data   | Model too simple, too few features            |
| Diagnosis        | Train error $\ll$ test error             | Train error $\approx$ test error (both high)  |
| Fix              | Regularization, more data, simpler model | Add features, use a more expressive model     |

## Real-World Example: Medical Diagnosis

**Underfitting:** A model that always predicts “no disease” (high bias) — misses actual cases.

**Overfitting:** A model that memorises specific patient records — fails with new patients.

**Goal:** A model that learns the *general* relationship between symptoms and disease.

# Inductive Bias: The Assumptions We Make

## Definition

**Inductive bias** is the set of assumptions a learning algorithm makes to generalise beyond the training data. Without bias, no learning is possible.

## Examples by Algorithm

- **Linear regression:** Assumes linear relationship between features and target
- **KNN:** Assumes similar inputs have similar outputs (smoothness)
- **Decision trees:** Assumes hierarchical feature relevance
- **SVM:** Assumes large-margin separation
- **Neural networks:** Assumes composition of functions

## No Free Lunch Theorem (Wolpert, 1996)

No single algorithm is best for all problems. The “right” algorithm depends on the problem’s underlying structure—which we encode as inductive bias.

## The Bias–Variance Decomposition

$$\mathbb{E}[(y - \hat{f}(x))^2] = \text{Bias}^2 + \text{Variance} + \sigma^2$$

*(exact for squared-error loss; a useful intuition otherwise)*

- **Bias:** Error from incorrect assumptions
- **Variance:** Error from sensitivity to training data
- **Noise:** Irreducible error from the problem itself

### The trade-off:

- High bias  $\rightarrow$  underfitting
- High variance  $\rightarrow$  overfitting
- Need to balance both

## Key Insight

Strong inductive bias  $\rightarrow$  learns faster, may be wrong  
Weak inductive bias  $\rightarrow$  more flexible, needs more data

## Ivy League Alignment

# Stanford CS229: The Gold Standard

## Course Philosophy (Andrew Ng)

*"Machine learning is the science of getting computers to act without being explicitly programmed. In the past decade, machine learning has given us self-driving cars, practical speech recognition, effective web search, and a vastly improved understanding of the human genome."*

— Stanford CS229 Course Description

### CS229 Coverage:

- Supervised learning: linear regression, logistic regression, SVMs, neural networks
- Unsupervised learning: clustering, PCA, factor analysis
- Learning theory: bias-variance, regularization
- Reinforcement learning: MDPs, value iteration, policy gradients

## CS229's First Lecture

Andrew Ng opens with the Mitchell definition, the spam filter example, and the three paradigms. Students then implement linear regression in the first homework.

## CSE 813 Alignment

This course follows CS229's **structure and philosophy**:

- Week 1: Introduction & foundations
- Weeks 2-7: Supervised learning
- Weeks 8-9: Unsupervised learning
- Weeks 10-11: Neural networks & deep learning
- Weeks 12-13: Advanced topics & ethics

## Also Aligned With:

- MIT 6.867: Statistical inference foundation
- CMU 10-701: PhD-level foundations
- UC Berkeley CS189: Mathematical rigour
- Harvard CS181: Applied perspective

## Student Assessment Pool

## Instructions

Select the single best answer. Answers are **bolded** for instructor reference; present without bold to students.

- Q1. Who provided the canonical formal definition of machine learning as “*a computer program is said to learn from experience  $E$  with respect to some class of tasks  $T$  and performance measure  $P$ ...*”?
- ☐ (a) Arthur Samuel    (b) **Tom Mitchell**    (c) Geoffrey Hinton    (d) Yann LeCun
- Q2. Which paradigm of machine learning uses **unlabeled data** to discover hidden structure?
- ☐ (a) Supervised learning    (b) **Unsupervised learning**    (c) Reinforcement learning    (d) Semi-supervised learning

## Instructor Notes

Q1: Students commonly confuse Samuel (who coined the term) with Mitchell (who gave the formal definition).  
Q2: Reinforcement learning also uses unlabeled data but through interaction; the key distinction is the presence of rewards.



- Q3. A model that performs well on training data but poorly on test data is suffering from:
- ☐ (a) Underfitting
  - ☒ (b) **Overfitting**
  - ☐ (c) High bias
  - ☐ (d) Data leakage
- Q4. Which of the following is an example of a **supervised learning task**?
- ☐ (a) Customer segmentation
  - ☐ (b) Anomaly detection
  - ☒ (c) **House price prediction**
  - ☐ (d) Market basket analysis

### Instructor Notes

**Q3:** Overfitting is the classic case of high training accuracy + low test accuracy.

**Q4:** Customer segmentation, anomaly detection, and market basket analysis are unsupervised learning tasks (no labels available).

## Instructions

State True or False. Award full marks only if the student provides a **one-sentence justification**.

| Statement                                                                           | Answer |
|-------------------------------------------------------------------------------------|--------|
| 1. Reinforcement learning requires labeled training examples.                       | False  |
| 2. The goal of machine learning is to perform well on unseen data (generalization). | True   |
| 3. Spam filtering is an example of unsupervised learning.                           | False  |
| 4. Inductive bias refers to the assumptions a learner makes in order to generalize. | True   |

## Justification Model Answers

- 1: RL uses rewards, not labels; it learns through trial and error.
- 3: Spam filtering uses labeled emails (spam/ham), making it supervised learning.
- 4: Inductive bias is the set of assumptions that enables generalization beyond training data.

## Instructions

Answer each question in **100–150 words**. Use specific terminology covered in class.

- Q1. Define machine learning using Tom Mitchell's formal definition. Explain each component (Task T, Experience E, Performance P) with a concrete example.
- Q2. Differentiate between **supervised**, **unsupervised**, and **reinforcement learning**. Provide one real-world application for each.
- Q3. What is the difference between **overfitting** and **underfitting**? How does each affect model generalization?

## Instructions

Answer in **200–300 words**. Show step-by-step reasoning. Partial credit is awarded for correct reasoning even with incomplete conclusions.

### A1. Problem Formulation (Healthcare):

A hospital wants to build a system to predict patient readmission risk within 30 days of discharge.

- Formulate this as a machine learning problem using Mitchell's **T-P-E** framework.
- What type of learning paradigm is this?
- What challenges might arise in this healthcare application?

### A2. Bias-Variance Analysis:

Compare the inductive bias of a linear model vs. a decision tree.

- How does each learner's bias affect its ability to generalize?
- In what scenarios would each be preferred?
- How do you diagnose whether bias or variance is the dominant problem?

# Textbook & Reading References

## Primary Textbooks for Week 1

- **[Mitchell]** Mitchell, T. M. (1997). *Machine Learning*. McGraw-Hill. **Chapter 1**
- **[Bishop]** Bishop, C. M. (2006). *Pattern Recognition and Machine Learning*. Springer. **Chapter 1**
- **[Murphy]** Murphy, K. P. (2012). *Machine Learning: A Probabilistic Perspective*. MIT Press. **Chapter 1**
- **[Hastie]** Hastie, T., Tibshirani, R., & Friedman, J. (2009). *The Elements of Statistical Learning*. Springer. **Chapters 1, 2**
- **[Shalev-Shwartz]** Shalev-Shwartz, S., & Ben-David, S. (2014). *Understanding Machine Learning*. Cambridge. **Chapters 1, 2**

## Supplementary—Covered in Week 1 Part 2 (Research Articles)

- Samuel, A. L. (1959). *Some Studies in Machine Learning Using the Game of Checkers*. IBM Journal. **[PRIMARY]**
- Jordan, M. I., & Mitchell, T. M. (2015). *Machine Learning: Trends, Perspectives, and Prospects*. Science, 349(6245). **[SURVEY]**
- Wolpert, D. H. (1996). *The Lack of A Priori Distinctions Between Learning Algorithms*. Neural Computation. **[NO FREE LUNCH]**

# Questions & Discussion

Week 1, Part 1 | CSE 813 Machine Learning

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*“Machine learning is the science of getting computers to act without being explicitly programmed.”*

— Andrew Ng

Next Lecture: Week 1, Part 2

Research Articles: Samuel (1959) · Jordan & Mitchell (2015)

Case Study: Spam Filtering (Classic ML Success Story)